Detecting Infection in Corn Leaves Using YOLO26-Based Density Peaks Clustering

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ABSTRACT

Corn is widely used as a food and renewable energy source. Due to its sustainability, corn is used to produce various chemicals and serves as a popular substitute. However, maize leaf infection can reduce corn productivity. This study presents a novel method, YOLO26-SDW-DPC, to quickly detect and classify infected leaves. The YOLO26-SDW-DPC algorithm can be considered a replacement for stride convolution, using a spatial Distance-based Density Peaks Clustering (DPC) in the backbone to improve usability and adaptability. In addition, a deformable attention–based vision transformer is used.

Keywords-corn leaf infection identification; DPC-Conv; deformable attention–based vision transformer; density peaks clustering; YOLO26

I. INTRODUCTION

Corn is widely grown and used as a primary food source among many people around the world. In addition, it is widely utilized as an industrial raw material to produce synthetic polymers, chemicals, biofuels, and others [1]. For many farmers, corn production is a major source of income. In China, corn serves different industrial purposes as a Renewable Energy Source (RES), in addition to being essential to maintain national food security [2]. However, corn leaf disease is widespread, drastically reducing total harvests, and is the primary cause of low maize production. Many deep learning methods have been proposed to identify maize leaf diseases and mitigate their impact on corn productivity [3, 4].

Deep learning techniques are used in various fields [5] due its the ability to incorporate features across multiple layers to deliver cutting-edge outcomes [6, 7]. The main aim of object detection in image analysis is to accurately identify objects in images or videos. Famous algorithms for object detection include YOLO, SSD, and R-CNN [8]. In [9], YOLO-based deep learning models were used to detect healthy, stressed, and damaged plants in a precision agriculture system, while in [10], an efficient YOLOv8 approach was used to classify 32 diseases and predict crop losses early. In [11], the YOLO-SPDNet model was used to classify tomato diseases, employing Attention Mechanisms (AMs) to improve localization performance.

A. Phases in Deep Learning-Based Object Detection

- Conventional object detection: In the 2000s, many researchers worked on feature extraction and classification using normal classification algorithms. These approaches had many limitations and drawbacks, such as difficulty in feature selection and improving classifier performance, high complexity, and the need for regular human intervention.
- Convolutional Neural Networks (CNNs): The advancement of CNNs led researchers to explore their implementation in object detection, which has become one of the most popular techniques.
- One-stage target detection: Unlike multi-stage detection frameworks, one-stage detectors, such as SSD and YOLO, are designed to achieve faster inference by eliminating multiple processing stages. However, although this design can improve speed, it can also reduce detection precision.
- Transformer-based object detection: Initially developed for Natural Language Processing (NLP), the concept of transformers has been recently applied to object detection due to its capacity for parallel processing and effective feature modeling. These methods rely on a concept called self-attention to analyze image data, thereby minimizing dependence on conventional convolutional architectures. Thus, transformer-based methods exhibit better efficiency and performance while reducing computational cost.

B. Corn/Maize Disease Detection

In [12], a single-stage, multi-scale-based feature fusion method was proposed to detect maize leaf diseases using YOLOv5. Weights were tuned to features to increase their significance, and an AM was implemented to enhance the network's capacity for feature extraction and learning. Experiments were conducted under challenging conditions, including sparse distributions and overlapping targets, with the MFF-CNN model outperforming in both detection time and accuracy. Convolutional procedures have limited capacity to represent images globally, leading to inadequate extraction of global contextual information. To overcome these limitations, DropBlock convolutional regularization was applied to improve detection accuracy, while transformer-based components were added to strengthen the baseline architecture. In [13], a lightweight YOLOv4-tiny framework was introduced for efficient weed removal in corn fields, achieving an mAP of 81.89%. Another study [14] attempted to address problems such as low accuracy and outdated technology in the identification of corn leaf diseases.

In recent years, research on corn disease detection has advanced rapidly, driven by deep learning–based object detection techniques. This study utilizes a lightweight YOLO-based detection framework for the rapid identification of corn leaves. The main contributions of this study are:

- Different preprocessing approaches to image resizing and normalization.
- Data augmentation, including random horizontal flipping and resizing, to improve generalization.
- A novel model, based on YOLO26-SDW-DPC, to detect infections in corn leaves.
- A Deformable Attention-Transformer model to classify the images as healthy or infected.

II. METHODOLOGY

A. Datasets

Corn images were used in this work, consisting of both healthy and infected maize leaves, from a publicly available dataset [15].

B. Healthy and Infected Images

The dataset [15] comprises images of healthy and infected corn leaves. This study used a total of 10060 maize leaf images, of which 4050 were infected and 6010 were healthy. The input images were split into training (70%), validation (20%), and test (10%) sets, respectively. Table I provides details on the corn dataset and the data split.

TABLE I. CORN IMAGE DATASET

Category	Total	Training	Testing	Validation
Infected Images	4050	2835	405	810
Healthy Images	6010	4207	601	1202
Total Images	10060	7042	1006	2012

C. Preprocessing

Five methods were used for resizing. The first is an image optimizer method that resizes images from 4608×3456 to 1200×900 pixels, and the data pipeline performs another scaling step, standardizing the image resolution from 1200×900 to 224×224, which is appropriate for training deep-learning networks.

D. Data Augmentation

Data augmentation prevents overfitting by growing the dataset size. The following consecutive transformations were applied to the images used for training: Random horizontal rotation of up to 15°, random horizontal flipping, random resizing to 256×256, center cropping to 224×224, and normalization. Normalization was performed after the testing and validation datasets were scaled to 256×256, and the center was trimmed to obtain a 224×224 image resolution.

III. YOLO26 BASED MODEL

Based on the number of parameters, YOLO26 is divided into five versions: Nano, Small, Medium, Large, and Extra Large [16-18]. The larger models in this series have the slowest detection speed but perform well in terms of accuracy. All five variants have identical network topologies; the primary distinctions lie in network size and detection speed. The two stages of feature processing are spatial pyramid pooling and multi-scale fusion. Various feature map stages are employed for up-sampling and fusion procedures. Spatial pyramid pooling is used to further integrate features and expand receptive fields. The loss functions for the prediction object, category labeling, and bounding box prediction are adjusted to enhance detection performance.

The procedure can be described as follows: Images are subjected to mosaic data augmentation. Conv, bottleneck CSP, and SPPF structures are used in the Backbone section to extract features from the input image [19]. In addition to fully integrating low-level and high-level features, this feature fusion procedure preserves additional information about object positions [20]. Three detection branches are included in the prediction to identify targets of various sizes. Parameter weights are optimized using classification and boundary-box functions. The key novelty of YOLO26 is a lightweight, end-to-end architecture that significantly reduces the computational cost of detecting infections in corn leaves.

A. YOLO26-SDW

This work is based on YOLO26 [21] using the Squared-Distance-Weighted (SDW) method, employing the Wise-IoU loss function, adding a deformable AM, and using downsampling convolution modules. Figure 1 shows the YOLO26 model for corn image classification. This model eliminates non-maximum suppression, reduces the parameter count to 26 million, introduces ProgLoss and STAL, optimizes the bounding-box loss on the training data, and adds a deformable attention model. Algorithm 1 describes the proposed YOLO26-SDW method for detecting corn leaf diseases.

```
Algorithm 1: YOLO26-SDW for detecting
infection in corn leaves
Input: Corn image dataset
Hyperparameters: Number of Layers,
        batch_size, Function
Output: Trained YOLO26-SDW model
Step 1: Image Dataset
Step 2: Preprocessing
Step 3: Data Augmentation
Step 4: Load the YOLO26-SDW model,
        configure the detection head for
        small objects
        Initialize the model to fixed
        input size
        Apply hyperparameters
        Feature maps of scale values X (1)
Step 5: Compute images using Attention
        Matrix (2)
```

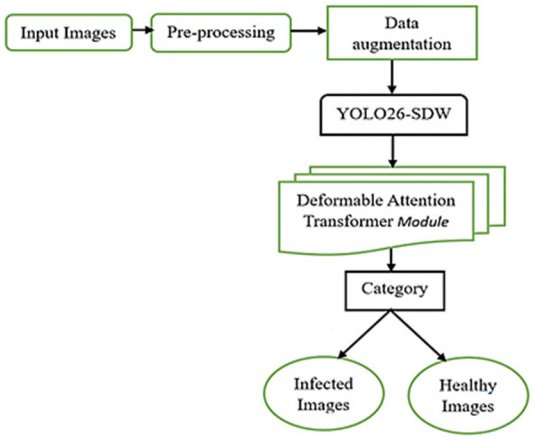


Fig. 1. YOLO26 model for corn image classification.

B. Dense Pyramid Convolution (DPC) Module

DPC extracts multiple features from the image using dense connections and is mostly used for image classification and segmentation of different objects. Conventional stride convolution may result in poor feature representation learning and loss of fine-grained image information. DPC-Conv improves a CNN's capacity to learn image features by replacing strided convolutions and pooling layers. DPC-Conv is a spatial encoding method that transforms the spatial representations of feature images into depth information. This module consists of a non-stride convolutional layer and a spatial-to-depth DPC layer [22]. The feature value map function of the dimension $S \times S \times C_1$ is sliced by the spatial-to-depth layer model.

Four subfeature maps of shape $(\frac{S}{2}, \frac{S}{2}, C_1)$ are produced when $scale = 2$. The channel dimension is created by converting the feature space dimension, and grows from C_1 to $4C_1$, as the spatial information dimension decreases by a factor of 2. Feature maps from different channels are initially grouped and integrated at the channel level. Once the fusion is completed, the feature map is combined with the other available feature maps. Then, it is passed through a stride convolution, which can maintain the original spatial resolution while reducing the number of channels to half of the original.

$$f_{0,0} = X[0{:}S{:}scale, 0{:}S{:}scale]$$

$$f_{1,0} = X[1{:}S{:}scale, 0{:}S{:}scale]$$

$$f_{scale-1,1} = X[scale-1{:}S{:}S,\ 1{:}S{:}scale]$$

$$f_{0,\ scale-1} = X[0{:}S{:}scale-1,\ S{:}scale]$$

$$f_{scale-1,scale-2} = X(scale-1; S{:}scale,\ scale-1{:}S{:}scale) \quad (1)$$

C. Deformable Attention Transformer Module (DATM)

Transformers are well known for their ability to model long-range dependencies, support parallel computation, and provide interpretable representations. In recent years, their usage has increased considerably, demonstrating strong potential in different fields [23]. In [24], Vision Transformer (ViT)-based transfer learning methods were used for early detection of bean leaf diseases. A transformer architecture includes two main components: an encoder and a decoder. The input representation, denoted X, is passed through multiple encoder layers to generate an encoded feature matrix. This encoded representation is then processed by the decoder, which applies masking mechanisms to enable sequential prediction. The DAT model is a self-Attention-based model that mainly focuses on reducing computational cost and improving each model's accuracy. To extract the best feature, DAT uses encoder layers for the best feature of the image matrix.

Table II summarizes the architecture details for the models used in this study.

TABLE II. REPRESENTS A MODEL FOR TRAINING LAYERS

Methods	Number of layers	Parameters	Batch size	Function
AlexNet	8	20 million	9	Pooling
ResNet50	40	24 million	15	Dense
VGG-16	16	14 million	19	FC2
MAX-ViT	12	11 million	20	Soft-max
YOLO-26 with DATM	20	26 million	32	ProgLoss

Multi-Head Attention, Add, Norm, and Feed Forward make up the encoder and decoder blocks. Multi-Head Attention consists of several self-attention systems. After obtaining matrices Q, K, and V through linear transformation of the input matrix, the output matrix is calculated using Multi-Head Attention, represented by a weight:

$$Multi\ Head\ Attention(Q, K, V) = \text{content}(head_1, head_2, \ldots, head_n) \quad (2)$$

where $head_i = Attention\ Matrix = QW_i^{\ j}, KW_i^{\ k}, VW_i^{\ u})$, and $(head_1, head_2, \ldots . head_n)$ represents n heads. The number of weights is denoted by n, and the mapping values are Q, K, and V.

For training multi-layer networks, Add and Norm are residual connections.

$$LayerNorm\big(X + FeedForwardNetworkModel\ (X)\big) \quad (3)$$

LayerNorm is a normalization module that transforms each layer's neurons by subtracting their mean and dividing by their variance, thereby accelerating convergence. X is the input vector following the Feed Forward Network (FFN) or the Multi-Head Attention. Two fully connected layers make up the Feed Forward layer; the first layer has the ReLU activation function, while the second layer does not have an activation function.

$$FFN(x) = ReLU\ (xW_1 + b_1)W_2 + b_2 \quad (4)$$

The Transformer's Feed-Forward layer is essential for increasing network detection capacity and computational efficiency.

D. Vision Transformer with Deformable Attention (DAT)

The DAT's fundamental concepts include plug-and-play, dynamic sampling, and Deformable Attention (DA). The transformer's conventional AM is utilized for all Q values, focusing on global positional cues. Building on the Deformable Convolution Network (DCN), DATM integrates ViT with DCN. Every Q value learns offsets of the position, which are traits it inherits from the ViT and DCN [25] network structure, where it initially applies a 5×5 Conv operation to features given as inputs, then applies a GELU activation function, and lastly uses 1×1 convolution to produce a two-dimensional output offset. Following the modification, the interpolation looks like q, k_1, v_1 and the relative position offset R undergoes the Multi-Head AM before being projected through W_0 to produce the output matrix z. k_1 and v_1 stand for the deformed K and V, respectively, for the interpolation procedure:

$$q = xW_q, k_1,\ (x_1W_k = v_1W_k\ ) \quad (5)$$

IV. EXPERIMENTAL RESULTS AND ANALYSIS

The proposed corn leaf infection detection system was simulated using Python 3.7 on an i7 processor with 16.0 GB of RAM and an HD 4600 graphics card. The evaluation criteria involved a confusion matrix, accuracy, precision, recall, F1 score, and mAP.

$$Accuracy = \frac{TN+TP}{TN+TP+FP+FN} \quad (6)$$

$$Precision = \frac{TP}{TP+FP} \quad (7)$$

$$Recall = \frac{TP}{TP+FN} \quad (8)$$

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision+Recall} \quad (9)$$

$$AP = \frac{1}{n}\sum_{k=1}^{k} Precision\ (k) * Recall\ (k) \quad (10)$$

$$mAP = \frac{1}{n}\sum_{k=1}^{n} AP_k \quad (11)$$

The number of correctly predicted positive (TP) and negative (TN) samples indicates the accuracy of the prediction. FP denotes negative samples mispredicted as positive, and FN denotes positive samples mistakenly predicted as negative. Table III shows that the YOLO-SDW algorithm is suited for identifying maize leaf infections. However, despite augmentation, overfitting was noted, as the model achieved significantly better accuracy in both the training and validation sets. The proposed model reduces the parameter count to 26 million, introduces ProgLoss and STAL, optimizes the bounding-box loss on the training data, and adds a deformable AM to the Neck.

Figure 2 illustrates the accuracy and loss on the training and validation phases of the proposed model (YOLO26-SDW-DATM) over 100 epochs.

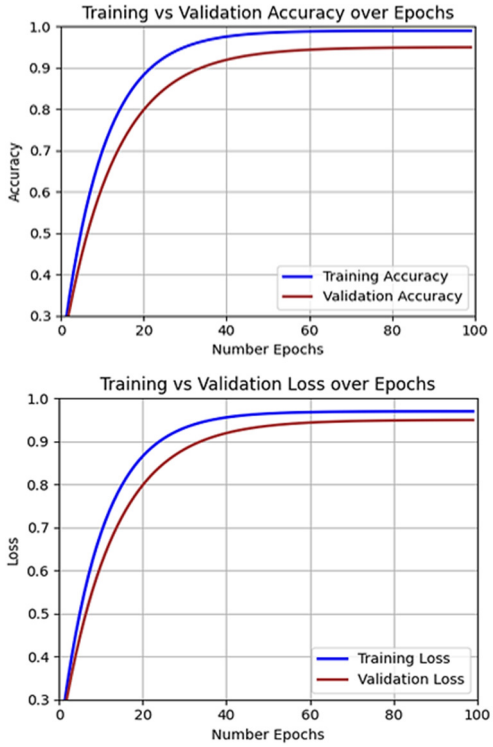


Fig. 2. Training and validation accuracy and loss for the proposed model.

TABLE III. COMPARISON OF DIFFERENT MODELS

Methods	Test Accuracy	Validation Accuracy	Training Accuracy	Precision	Recall	F1-score
AlexNet	93.48	92.47	95.91	96.17	93.47	91.84
Resnet-101	91.71	89.73	93.75	96.87	91.49	94.10
ResNet50	87.97	89.98	92.76	93.76	90.73	92.22
VGG-16	88.79	89.27	83.79	91.57	89.76	90.65
MAX-VIT	95.48	92.79	94.57	93.76	90.72	92.21
YOLO-26 with DATM	88.71	94.12	97.76	87.14	86.54	86.83

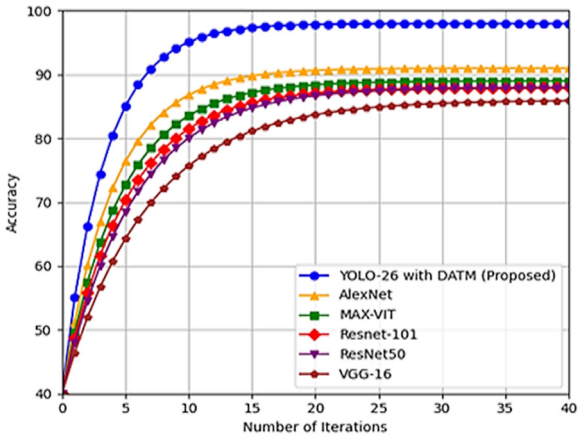


Fig. 3. Accuracy per epoch comparison across models.

Figure 3 compares a comparison of validation accuracy across epochs for the models tested. The proposed model achieved an initial accuracy of 40% in the starting iterations. After increasing the number of iterations to 40, the proposed model achieved 97% accuracy, surpassing that of other pre-trained models.

Figures 4 and 5 show testing and training confusion matrices, respectively, for the proposed model.

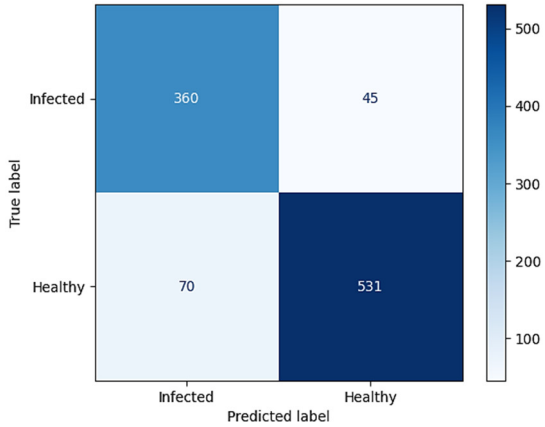


Fig. 4. Confusion matrix for test data on corn disease detection.

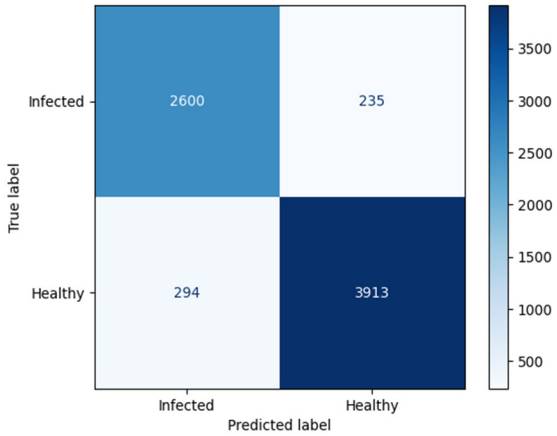


Fig. 5. Confusion matrix for training data on corn disease detection.



Fig. 6. YOLO26-SDW algorithm for corn disease detection.

The proposed YOLO26-SDW model introduces several improvements to address key issues in the detection of maize leaf diseases under real field conditions. These significantly improve detection performance, particularly for lesions with varying shapes and sizes. As a result, the model's ability to localize and classify diseased regions on maize leaves is increased. However, the model demonstrated overfitting, which remains to be addressed.

V. DISCUSSION

Although integrating DPC Conv and DAT improved training performance, it also increased the model's complexity, leading to greater computational requirements and the need for larger GPUs. This may limit practical deployment on resource-constrained platforms, such as edge computing, which have limited hardware capacity. In addition, the diversity of agricultural plant diseases and the emergence of new disease types may require repeated retraining and model tuning to fit new data. Model performance may also be affected by changes in background and illumination, which are common in real-world environments.

In summary, the proposed YOLO26-SDW method is promising for identifying maize leaf diseases. Future research can explore in depth areas such as enhancing the DAT mechanism to accommodate additional plant disease detection methods to increase the model's resilience in real-world settings. Next, more advanced data augmentation methods can be used during the training phase, such as simulating different lighting conditions for the leaves and background to improve testing performance. Furthermore, it is very important to investigate the development of a more lightweight YOLO-SDW method for use on edge computing devices.

VI. CONCLUSIONS

In real-world scenarios, the detection of corn leaf diseases is challenged by factors such as varying disease symptoms, partial leaf occlusion, and interference from surrounding objects, including weeds. To resolve these issues, this study developed a YOLO26-SDW approach, built on the YOLO26 framework, that replaces conventional stride convolution with a spatial-depth-preserving convolution (DPC). This modification helps to retain fine-grained feature information and feature map representation by reducing the loss introduced by strided convolution and pooling operations.

The DAT combines the advantages of ViTs and DCN. By adding an offset prediction network, the model can adaptively shift keys and values toward more informative regions during training, thereby improving feature information while eliminating redundant computation. In addition, the Wise-IoU loss function was used instead of the conventional IoU loss. This loss function uses a dynamic weighting strategy that assigns lower gradient contributions to low-quality anchor boxes and adjusts the gradient scale based on outlier characteristics.

The proposed YOLO26-SDW algorithm shows promising performance, being a valuable contribution to advancing corn leaf disease detection and crop target recognition. These results provide a solid basis for further research and model refinement. Future work will aim to improve the robustness of the proposed approach for identifying maize leaf blight under complex, variable environmental conditions and investigate its adaptation for use in edge devices.

DECLARATION OF COMPETING INTERESTS

The authors declare no conflicting interests that could have influenced the results of this study.

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Not applicable.

DATA AVAILABILITY

The dataset used in this study is available at [15].

FUNDING SOURCES

Not applicable to this study.

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